Automated Data load Validation in Snowflake

--Step:-1---------Create Database and schema----------

```
CREATE OR REPLACE DATABASE AUTO_LOAD_DB;

CREATE OR REPLACE SCHEMA AUTO_LOAD_DB.PUBLIC;

USE DATABASE AUTO_LOAD_DB;
USE SCHEMA PUBLIC;
```

--Step:- 2----------Create Final Target Table----------(This table stores only clean/validated data)

```
CREATE OR REPLACE TABLE CUSTOMER_TARGET (
   CUSTOMER_ID INT,
   CUSTOMER_NAME STRING,
   EMAIL STRING,
   CITY STRING,
   CREATED_AT DATE
);
```

--Step:- 3----------Create Audit Log Table_____(This table stores success/failure details)

```
CREATE OR REPLACE TABLE LOAD_AUDIT_LOG (
   LOG_ID INT AUTOINCREMENT,
   FILE_NAME STRING,
   LOAD_STATUS STRING,
   ROW_COUNT INT,
   ERROR_MESSAGE STRING,
   LOAD_TIME TIMESTAMP_NTZ DEFAULT CURRENT_TIMESTAMP()
);
```

Step:-4----------Create CSV File Format----------

```
CREATE OR REPLACE FILE FORMAT CSV_FILE_FORMAT
TYPE = CSV
FIELD_DELIMITER = ','
SKIP_HEADER = 1
FIELD_OPTIONALLY_ENCLOSED_BY = '"'
NULL_IF = ('NULL', 'null', '')
EMPTY_FIELD_AS_NULL = TRUE;
```

```
---Step 5: Create Snowflake Internal Stage------------------

CREATE OR REPLACE STAGE CUSTOMER_STAGE
FILE_FORMAT = CSV_FILE_FORMAT;

LIST @CUSTOMER_STAGE;

----------Step 7: Create Stored Procedure---------------
CREATE OR REPLACE PROCEDURE LOAD_VALIDATE_CUSTOMER(FILE_NAME STRING)
RETURNS STRING
LANGUAGE SQL
AS
$$
DECLARE
   NULL_COUNT INT DEFAULT 0;
   DUP_COUNT INT DEFAULT 0;
   TOTAL_ROWS INT DEFAULT 0;
   RESULT_MSG STRING;
   COPY_SQL STRING;
BEGIN

   CREATE OR REPLACE TEMPORARY TABLE TEMP_CUSTOMER_LOAD (
      CUSTOMER_ID INT,
      CUSTOMER_NAME STRING,
      EMAIL STRING,
      CITY STRING,
      CREATED_AT DATE
   );

   COPY_SQL :=
      'COPY INTO TEMP_CUSTOMER_LOAD
       FROM @CUSTOMER_STAGE
       FILES = (''' || FILE_NAME || ''')
       FILE_FORMAT = (FORMAT_NAME = CSV_FILE_FORMAT)
       ON_ERROR = "CONTINUE"';

   EXECUTE IMMEDIATE :COPY_SQL;

   SELECT COUNT(*) INTO :TOTAL_ROWS
   FROM TEMP_CUSTOMER_LOAD;

   SELECT COUNT(*) INTO :NULL_COUNT
   FROM TEMP_CUSTOMER_LOAD
```

```
    WHERE CUSTOMER_ID IS NULL
       OR CUSTOMER_NAME IS NULL
       OR EMAIL IS NULL;

    SELECT COUNT(*) INTO :DUP_COUNT
    FROM (
        SELECT CUSTOMER_ID
        FROM TEMP_CUSTOMER_LOAD
        WHERE CUSTOMER_ID IS NOT NULL
        GROUP BY CUSTOMER_ID
        HAVING COUNT(*) > 1
    );

    IF (:NULL_COUNT = 0 AND :DUP_COUNT = 0) THEN

        INSERT INTO CUSTOMER_TARGET
        SELECT CUSTOMER_ID, CUSTOMER_NAME, EMAIL, CITY, CREATED_AT
        FROM TEMP_CUSTOMER_LOAD;

        INSERT INTO LOAD_AUDIT_LOG
        (
            FILE_NAME,
            LOAD_STATUS,
            ROW_COUNT,
            ERROR_MESSAGE
        )
        VALUES
        (
            :FILE_NAME,
            'SUCCESS',
            :TOTAL_ROWS,
            NULL
        );

        RESULT_MSG := 'SUCCESS: File loaded and validated successfully. Rows inserted: ' ||
:TOTAL_ROWS;

    ELSE

        INSERT INTO LOAD_AUDIT_LOG
        (
            FILE_NAME,
            LOAD_STATUS,
            ROW_COUNT,
```

```
        ERROR_MESSAGE
    )
    VALUES
    (
        :FILE_NAME,
        'FAILED',
        :TOTAL_ROWS,
        'Validation failed. Null count: ' || :NULL_COUNT || ', Duplicate count: ' || :DUP_COUNT
    );

    RESULT_MSG := 'FAILED: Validation failed. Null count: ' || :NULL_COUNT || ', Duplicate
count: ' || :DUP_COUNT;

  END IF;

  RETURN RESULT_MSG;

EXCEPTION
  WHEN OTHER THEN

    INSERT INTO LOAD_AUDIT_LOG
    (
        FILE_NAME,
        LOAD_STATUS,
        ROW_COUNT,
        ERROR_MESSAGE
    )
    VALUES
    (
        :FILE_NAME,
        'FAILED',
        0,
        'Unexpected error occurred during load'
    );

    RETURN 'FAILED: Unexpected error occurred during load';

END;
$$;

LIST @CUSTOMER_STAGE;
```

----------Step 8: Test with Valid CSV File----------

```
CALL LOAD_VALIDATE_CUSTOMER('customer_valid_20_records.csv');


SELECT * FROM CUSTOMER_TARGET;

SELECT * FROM LOAD_AUDIT_LOG
ORDER BY LOAD_TIME DESC;
```

----------Step 9: Test with Invalid CSV File (Now call the invalid file)----------

```
CALL LOAD_VALIDATE_CUSTOMER('customer_invalid_20_records.csv');

SELECT * FROM LOAD_AUDIT_LOG
ORDER BY LOAD_TIME DESC;
```